

Measuring URLs in the Common Crawl index

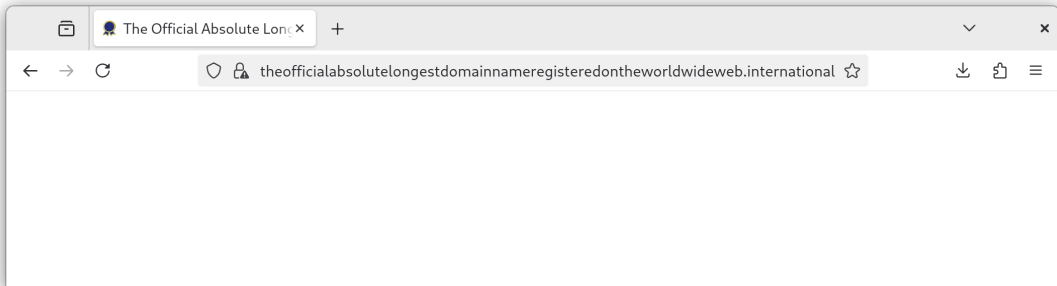
How long are most URLs?

2026-01-08

Pierre Marshall

1 Introduction

Motivation



Motivation (ii)



How much memory should we reserve for a url string?

```
String::with_capacity(?)
```

What are the limits?



Firefox > 300,000



Safari > 64,000



Chrome 32,779



Edge 2,047

What are the limits? (ii)



The 414 (URI Too Long) status code indicates that the server is refusing to service the request because the request-target is longer than the server is willing to interpret.

— RFC 7231 Section 6.5.12.

a free, open repository of web crawl data that can be used by anyone

- An archive?
- An index?

```
org,example)/index.html 20220106150849300 {  
  "url": "https://example.org/index.html",  
  "status": 200  
}
```

Method



Start with cc-index-paths, download and decompress each url in the file.

We then have a 5GB chunk of the index.

For each line in this chunk, truncate the first few bytes to get the json, then deserialise it using `serde nanoserde` to a struct which looks like this:

```
struct IndexRecordParsed {  
    url: String,  
    url_length: u16,  
    i18_url_length: u16,  
    status: u8,  
}
```

Method (ii)



Then push that to a Vec, and deduplicate at the end with

```
dedup_by(|a, b| a.url == b.url).
```

Iterate over the newly created Vec and map all these values to a comma-separated list. Write it out to file. Start again.

There are for sure better ways of doing this!

Performance



- Use all the threads
- Don't touch the file system
- Use lifetime annotations / scope your functions.
- (grug brain solution) get a bigger server
- Use lazy execution



1.g.a URL encoding

If the url `https://www.example.com/` is 24 characters long, how many characters graphemes does `https://www.example.com/讯` contain?

Methodology (ii)



1.g.b IDNA bonus round

`https://haurakicollective.māori.nz`

Further reading: Evan Hahn [1], Henri Sivonen [2]

2 Results

Charts



status code	mean	median	% of total
1xx	101	58	0%
2xx	74	63	85.6%
3xx	77	63	8.6%
4xx	69	58	5.5%
5xx	61	50	0.3%
total	74	63	

Table 1: Average graphemes per url



Charts (ii)

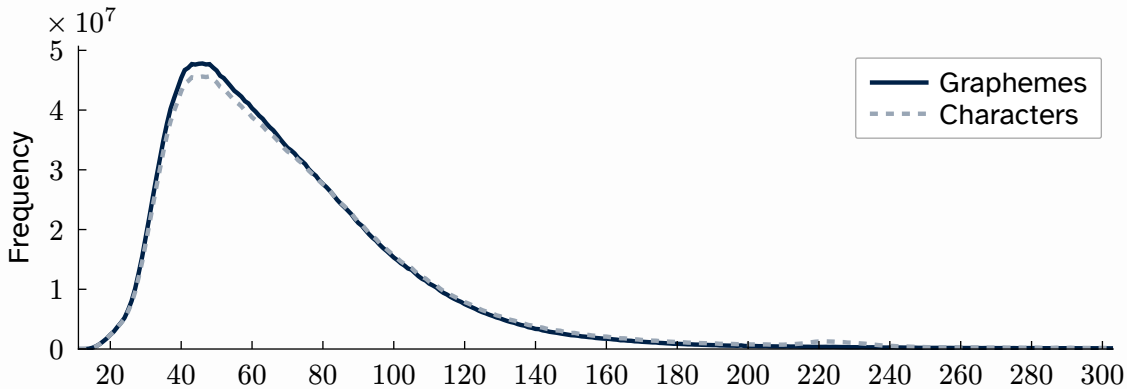


status code	mean	median
1xx	106	58
2xx	78	65
3xx	81	64
4xx	72	59
5xx	64	51
total	78	65

Table 2: Average characters per url



Charts (iii)



Quartile values



Quartile	Raw characters	Graphemes
75%	89	86
80%	97	93
90%	121	114
95%	154	136
99%	303	244

Table 3: quartile values

Resources



Figure 1: Get my code at <https://github.com/Digital-Scholarship-Oxford/cc-most-urls>

Resources (ii)



Icons by SuperTinyIcons (edent)

References

- [1] Evan Hahn, “Why does 🇺🇸 have a length of 7 in JavaScript?.” [Online]. Available: <https://hsivonen.fi/string-length/>
- [2] Henri Sivonen, “It’s Not Wrong that ♂.length == 7.” [Online]. Available: <https://hsivonen.fi/string-length/>
- [3] “Introduction to Common Crawl Datasets,” in *Getting Structured Data from the Internet: Running Web Crawlers/Scrapers on a Big Data Production Scale*, Apress, 2020, pp. 277–324.

Resources (iii)



- [4] Kelvin Tan, “Average length of a URL (Part 2).” [Online]. Available: <https://web.archive.org/web/20240114201737/https://www.supermind.org/blog/740/average-length-of-a-url-part-2>
- [5] Paul Dixon, “What is the maximum length of a URL in different browsers?.” [Online]. Available: <https://stackoverflow.com/questions/417142/what-is-the-maximum-length-of-a-url-in-different-browsers/417184#417184>